Drawbox: Research and Next Steps

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Two rounds of research on the front-opening cooler drawer idea: where it fits, what exists, what the market looks like, and the cheapest way to find out if people will buy it.

The short version

The idea holds up, but not where we first pointed it. The one place on a boat where a cooler is already trapped behind a seat, and where owners already pay to fix it, is under the leaning post on center consoles and bay boats. That's the first product. The passenger console on open-bow boats is the second, with a catch: on dual consoles over 21 feet that console is the bathroom, so the real target is the hollow box under the console seat on center console bowriders and jet boats. Pontoons are the biggest segment and the worst fit: the seat bases are 11 inches tall, which leaves room for a drinks drawer, not an ice chest.

Nothing like this is sold today. The closest things are $300 slide trays that roll a regular cooler out from under the leaning post, and $600 to $3,600 powered drawer fridges. The concept was patented in 1990 and let lapse, and four more individual inventors have patented versions since without shipping one. That's a warning about unit economics, not proof the idea is bad.

The call: build the leaning post version first as a kit around an off-the-shelf insulated tub, so there's no tooling money at risk, and run three cheap tests before spending anything real. Tilt-out is out in every version; a tub of meltwater tips, and cheap tip-out hinges are rated for sponges.

Where boats hide the cooler

The question that matters is not "is there dead space under the seat" but "where is a cooler already stuck behind a vertical face." Only two spots pass that test, and no boat builder publishes the dimensions of either.

Boat type	Where the cooler goes today	How you get at it	What we know about the size	Fit
Center console / bay boat	Loose cooler under the leaning post; small box under the forward console seat	Pull the whole cooler out, often on a $300 slide, then open the lid	28.5 to 29.5 in between post legs (Key West 203FS); 45 to 94 qt coolers fit; console box holds 28 to 40 qt	Primary target
Center console bowrider (Key West 210BR, Tidewater 230)	Hollow, uninsulated box under the forward console seat	Pop a snap-on cushion	Unmeasured; owners foam it to 1 in and say ice lasts an hour to a day	Strong fit, cavity is open and the complaints match
Jet boat (Yamaha AR, SX, FSH)	Port console locker and under-seat bay	Lockable console door; lift seat	Under-seat bay about 25 W x 18 D x 16 H in; a 40 qt fits, a 26.5 in wide cooler does not	Good fit
Dual console under 21 ft (Robalo R207)	Port console is a plain locker	Corner door with a strap	Walkway between consoles is 19 in	Possible, must open toward the cockpit
Dual console 21 to 25 ft (Grady Freedom, Whaler Vantage, Robalo R247, Pursuit, Scout)	Port console is the enclosed head; coolers live in the aft utility center or under bow seats	Lockable door or whole face swings open	Factory capacities only (35 qt Yeti pocket, 56 to 160 qt boxes)	No for the console
Sterndrive bowrider / deck boat (Sea Ray, Chaparral)	Carry-on cooler clips in under a bow seat	Lift cushion	Nothing published	No, every seat is already a top loader

Boat type	Where the cooler goes today	How you get at it	What we know about the size	Fit
Wake / ski boat	Built-in boxes are being dropped (Nautique 200 has none)	Lift cushion	Nothing published	No
Pontoon seat base	Uninsulated liner with a drain plug (Bennington); in-floor box (Barletta)	Lift the cushion, often blocked by a filler seat or table	Aftermarket bases 27 to 55 in wide, 19 to 22.5 in deep, 11 in tall; OEM heights unpublished	Drinks drawer only, defer
Pontoon helm stand	Open compartment behind a door; owners slide soft coolers in	Side or front door	37 x 23 x 34.5 in shell; amps and wiring eat the space on premium boats	Dealer-fit niche

The leaning post wins on three counts. The cavity is the same shape on every boat because it's sized around a loose cooler. There is already a $300 aftermarket category whose only job is front access. And it sits on an open deck, clear of the flotation foam that USCG rules put along the hull sides on boats under 20 feet, which is exactly where bowrider bow seats are.

What's already out there

Three adjacent categories and an empty middle. The real competitor is not a product, it's the cooler the buyer already owns plus a $300 slide.

Product	What it is	Price	Why it matters
Engel, Boat Outfitters, Birdsall cooler slides	Tray on slides that rolls a regular cooler out from under a leaning post	$300 to $320, plus the cooler	Proves owners pay for front access; sets the price anchor
DIY glides	Trailer bunk poly strips under the cooler	$3 to $25	The cheap end of the same fix

Product	What it is	Price	Why it matters
Dometic CoolMatic CD30	12V compressor drawer fridge, 28 L, under 10 in tall, 41 lb	$600 to $1,285	Shows builders design front-opening cold boxes for under-seat slots
Isotherm Drawer 130, Vitrifrigo DW180	Larger 12V drawer fridges, 40 to 135 lb	$2,225 to $3,610	Out of our price band entirely
Ninja FrostVault	Top-loading cooler with a small slide-out dry drawer	$200 to $349	SharkNinja kept the ice on top and used the drawer for dry goods, sidestepping meltwater and cold loss
DPI Marine tilt-out trash can	Cutout-mounted polymer bin, 18 x 16 in, sold at Overton's	Low	Proves boaters accept a cutout product in a seat base or console

Patents. Baltzell patented essentially this product (seat on top, insulated drawer out the front) in 1990 as US 5,100,198 and let it lapse in 1996. Chaney (2010), Beland (2014), McBeth (2017, active to 2036) and Kravchenko (2021) each patented a drawer-cooler variant as individuals, with no company behind them and no product. The live, funded patents are SharkNinja's 2024 pair, US12129099B1 and US12163734B1, "Insulated container with a drawer." A drawer that is itself the insulated vessel mounted into a boat seat is structurally different from a drawer inside a chest, but those claims need a read before any tooling is quoted. The pattern of individuals inventing and abandoning this for 35 years while slides became a category is the thing to take seriously.

Market size

The top of the funnel is big and the slice we can reach is narrow. None of the public numbers isolate "center console owner with a cooler under the leaning post," so the honest sizing is bottom-up from what people already pay.

Number	Value	Source
Registered boats in the US	About 11.8 million (2024)	NMMA
New powerboats sold, 2025	156,754	NMMA

Number	Value	Source
Pontoon share of new powerboats	33.6 percent, about 52,700 boats	NMMA
Center console and bay boat unit count	Behind NMMA's $2,525 paywall, not captured	NMMA
Marine aftermarket accessories spend	$12.4 billion (2024), within $55.6 billion total boating spend	NMMA
Rotomolded cooler market	About $1.8 billion (2025), about 12 percent tied to marine use	Vendor estimate
Yeti Tundra 45	$325, roughly $7 to $8 per quart	Yeti
Premium cooler range	$299 to $899	Retail
What a Key West dealer charges for cooler plus slide	$462, which owners call "pretty steep" and buy anyway	Key West owner forum

Pricing. A drawer that replaces both the slide and the loose cooler can ask $350 to $450. That sits below slide plus Yeti ($300 plus $325) and above slide plus Igloo. Per quart it lands in the premium band without a Yeti-grade ice story, so the pitch is access and a clean deck, not ice life.

Channel. Small marine outfitters reach buyers through direct web, Overton's, West Marine, and boat-magazine how-to coverage. That's the shelf this belongs on.

The cautionary tale. Coolest Cooler raised $13 million from 62,642 Kickstarter backers at $185, then found the product cost more to make and ship than the pledge price, got hit by tariffs going from 10 to 25 percent, and left 20,000 backers with nothing. Price after a real bill of materials and a tariff assumption, and stay off custom molds until the kit version has sold.

The three designs

One carriage, one slide set, one face. Three housings sized to the three cavities, built in order of how sure we are about the dimensions.

	Post	Console	Base
Goes in	Under a leaning post	Console seat box on open-bow boats	Pontoon seat base

	Post	Console	Base
Outside size	26 W x 14 D x 15 H in	22 W x 16 D x 13 H in	24 W x 18 D x 9 H in
Holds	About 45 qt, 48 cans plus ice	About 32 qt, 36 cans plus ice	48 cans in one layer, no loose ice
Cold source	Ice	Ice	Frozen gel floor plus pre-chilled cans
Opens toward	Front or aft under the post	Into the bow on a center console bowrider, toward the cockpit on a dual console, never into a 19 in walkway	Forward, into the walkway in front of the bench
Mount	Bolts to post legs and deck, no cutting	Cutout plus face frame	Cutout plus face frame, or a whole replacement rotomolded base
Who it fits	19 to 26 ft center consoles and bay boats with 24 in between the legs	Key West 210BR and 239FS, Tidewater 230, Grady Fisherman 216, Robalo R207, Yamaha jet boats	Any pontoon with standard 27, 36 or 50 in bases, if enough owners want it
Target price	$349 to $449	$299 to $399	$249 to $329
Build order	First. Proves the mechanism and the ice-loss number	Second. Same carriage, shorter tub, adds an airflow baffle	Third. Only after OEM base heights are measured and a gel floor prototype holds temperature

Why Post is first. The cavity is defined by the market already (the cooler slide makers publish the band: 17 to 27 in wide, 11.5 to 14.5 in deep, up to 54 qt), and it can be built with zero tooling: a stainless or HDPE carriage on $80 slides holding a $100 Engel 30 qt drybox.

Why Base is a different animal. Take an 11 in base, subtract the floor rails and the lid frame, and you get a 9 in drawer body with 7 in inside. That's 48 cans standing up with 2 in of headroom, and any ice you add either buries the cans or sits uselessly on top. Dometic

reached the same conclusion from the other direction: their under-10-inch drawer is a compressor fridge. If a Bennington, Harris or Barletta lounge measures 15 in or taller, Base inherits the Console body and the order flips.

Engineering

The hardware is solved; the open question is physics nobody has measured.

- **Slides.** Fulterer FR5609 20 in stainless full-extension slides are rated 220 to 250 lb per pair with both lock-in and lock-out, $79.62 a pair. A 25 qt load of cans and ice is 30 to 35 lb, 52 lb if packed solid, so a 220 lb pair is a 4x to 5x margin. Lock-in matters as much as lock-out: a loaded drawer without a positive latch walks open in chop. No slide maker publishes a salt-spray rating, so that's a supplier question.
- **Insulation.** Yeti and Pelican use about 2 in of foam per wall. In a 24 hour test with 20 lb of ice, a $109 Igloo and a $400 Yeti melted exactly the same amount. For a day-use boat drawer, 1 in walls are right; thicker walls just eat the cavity.
- **The unknown.** Every pull exposes the whole top of the ice to the air. Nobody has published cold loss per opening for a front-opening ice drawer, because none exists. A Key West owner found that air moving under the seat while underway killed ice faster than the walls did, so the face needs a compression seal and the cavity needs a baffle. This number comes only from a prototype.
- **Drainage.** Never to the bilge; owners say it makes the boat smell like a sewer. The drawer carries a removable sealed tub with a plug in the floor, dumped over the side. A tilt-out pours meltwater at the lip the moment it opens, which is the main reason tilt is out.
- **Flotation foam.** On outboard boats under 20 ft the USCG puts passenger flotation along the hull sides under the gunwales, which is where bow seats are. Leaning posts sit on open deck and pontoons are exempt. Console seat boxes on the boats we found are hollow (owners pour foam into them themselves), but the install instructions keep a probe step before cutting.
- **Tooling.** Rotomolding tools run $2,000 to $40,000 each, economic at 1 to 15,000 units a year; injection tools $15,000 to $100,000 and up. An insulated drawer that seals against a housing likely needs two or three tools. The kit route around an off-the-shelf drybox keeps tooling at zero, with hardware landing near $200 to $250 at retail part prices, which is tight at a $350 to $450 price unless the tub is bought wholesale or the carriage is the only custom part.

Next steps

All of this runs under $500 and a few weekends. Nothing here commits us to a mold. Put a name next to each one.

- [ ] **Measure 20 to 30 leaning posts.** Deck-to-seat-bottom height, distance between the legs, and depth. A dealer lot afternoon or a post in the Key West, Tidewater and Sea Hunt owner forums. This decides whether Post is one size or two. Measure three pontoon lounges (a Bennington, a Harris, a Barletta) and two console seat boxes on the same trip.
- [ ] **Build the zero-tooling prototype.** An Engel 30 qt drybox in an HDPE or stainless carriage on a pair of FR5609 slides, with a face frame and a Southco-style latch. About $250 to $300 in parts. Install it under one leaning post.
- [ ] **Run the ice test.** Same drybox, same day, same number of openings: one in the drawer, one top-loaded on the deck. Log the melt. This is the number that decides whether we can honestly claim day-trip retention, and it's the one no one has.
- [ ] **Put the site live and run the $399 test.** Landing page with the waitlist and the measurement form, pushed to center console and bay boat owners in Florida and the Gulf through forum posts and a small paid test. Run the pontoon version as a control. If the center console page outperforms, the segment call is confirmed. If neither converts better than a $300 slide would, it's a slide with better marketing and we shelve it.
- [ ] **Read the SharkNinja claims.** US12129099B1 and US12163734B1, independent claims only. Costs nothing, do it before any tooling quote.
- [ ] **Settle the name.** Drawbox is a placeholder. Check the trademark and the domain before it's on a page strangers see.

What we still don't know, and how to find out

Unknown	Why it matters	How to get it
Deck-to-seat height under leaning posts	Sets the Post tub height; currently inferred at 18 to 22 in	Tape measure, 20 to 30 boats
Console seat box interior on a 210BR, 239FS, Tidewater 230, Fisherman 216	Sets the Console tub height and whether 22 in wide is safe	Tape measure, four boats

Unknown	Why it matters	How to get it
OEM pontoon lounge heights	Decides whether Base is a drinks drawer or inherits the Console body	Tape measure, three brands
Ice loss per opening	The product's main claim	Prototype test
Salt-spray rating on stainless slides	Warranty exposure	Ask Fulterer and Accuride
Center console unit sales	Market sizing	NMMA paid report, or skip and size bottom-up

Sources

The pages the numbers above came from. Owner forum threads are where most of the real dimensions and complaints live, since no builder publishes them.

- NMMA 2025 powerboat sales and pontoon share
- NMMA 2024 recreational boating spend
- Boating Magazine, installing a cooler slide
- Engel cooler slides
- Key West forum, leaning post dimensions
- Key West forum, 189FS leaning post cooler
- Key West forum, 210BR console cooler upgrade
- Tidewater owners, insulating the console seat cooler
- JetBoaters, Yamaha under-seat cooler bay
- BoatTEST, Robalo R207 (19 in walkway)
- Boating Magazine, Grady-White Freedom 235
- Club Bennington, built-in cooler thread
- Club Bennington, storing ice chests
- Wise 36 in pontoon seat base, 11 in tall
- Dometic CoolMatic CD30
- Fulterer FR5609 stainless slide
- Engel 30 qt drybox
- Pew Pew Tactical, Igloo vs Yeti ice test

- USCG flotation guideline
- Baltzell 1990 seat cooler patent, US 5,100,198
- McBeth 2017 sliding drawer cooler, US 9,738,296
- GeekWire, Coolest Cooler shutdown
- Design News, when to choose rotomolding